MComp Project Presentation

Course-grained positioning of inanimate objects using IoT sensor data and machine learning

Overview

1. Introduction
2. Motivation
3. Background
4. Methodology
5. Experiments and Results
6. Limitations and Future Work
7. Conclusion

Introduction

Introduction: Topic and Hypothesis

Topic

Course-grained positioning of inanimate objects using IoT sensor data and machine learning

Motivating Questions

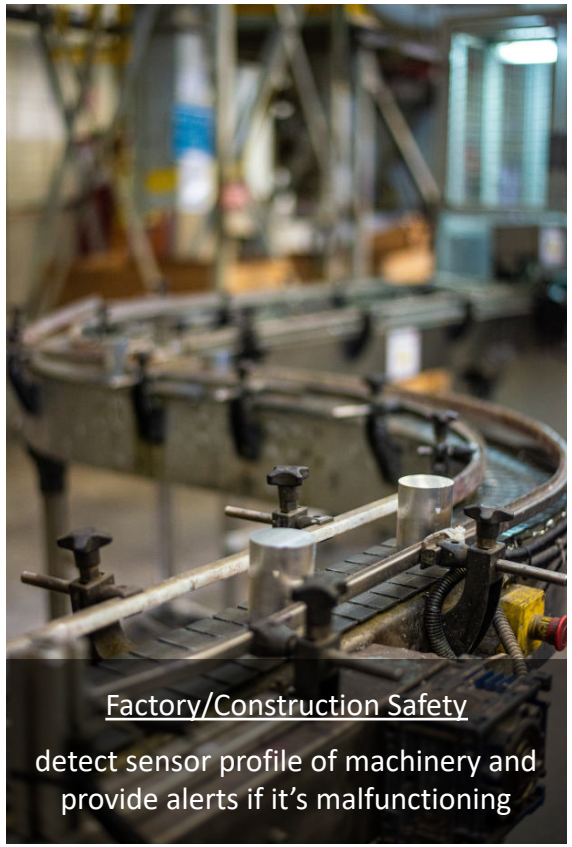
1. Can sensor data and machine learning identify what object is operating?
2. Can sensor data and machine learning identify approximate location of the object?
3. Can sensor data and machine learning identify the object's 3D location within the room or quadrant?

Introduction: Main Goals

1. Explore major applications for IoT systems: industrial safety, structural health monitoring, and fault detection
2. Demonstrate an end-to-end pipeline for processing IoT sensor data consisting of different sensor types and multiple nodes
3. Leverage IoT sensor data (specifically smartphone IMU sensors) and machine learning techniques
4. Evaluate effectiveness of different types of machine learning techniques for positioning of inanimate objects
5. Discuss limitations and future work

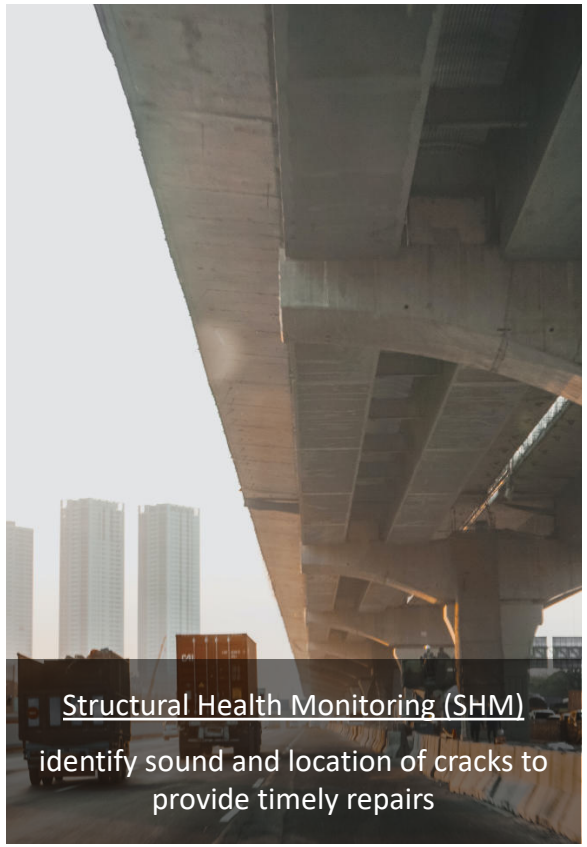
Motivation

Motivation



Factory/Construction Safety

detect sensor profile of machinery and provide alerts if it's malfunctioning



Structural Health Monitoring (SHM)

identify sound and location of cracks to provide timely repairs



General Fault Detection

recognize or predict equipment failure based on sensor readings

Background

Background: Localization Framework

Three Layered Challenge

Technology	Technique	Application
• IEEE 802.11 • RFID • UWB • Bluetooth • Visible light • Acoustics	• Received Signal Strength Indicator (RSSI) • Angle of Arrival (AoA) • Time of Flight (ToF) • Return Time of Flight (RTOF)	• Context Aware Location Based Marketing • Health Services • Disaster Management and Recovery • Security • Asset Management and Tracking

Two groups of localization approaches:

Fingerprinting: use pre-recorded radio map to infer locations using best matching

Propagation based: use RSSI values to calculate distances by computing path loss (high errors of up to 50%)

Background: Traditional Localization Methods

- Common localization techniques relies on ‘triangulation’
  - Angulation: Angle of Arrival (AOA) is used to determine angle
  - Lateration: Received Signal Strength (RSS) and Time of Flight (TOF) are used to determine distance from the node where the signal was picked-up
- Three methods above (AOA, RSS, and TOF) require precise clock synchronization
- Round-trip Time of Flight (RTOF) can be used to measure distance without clock synchronization
- Time Difference of Arrival (TDOA) can be used to measure the distance without synchronizing the clock between transmitter and receiver
- Other techniques beside triangulation include:
  - Proximity Detection (PD)
  - Map Matching (MM)

Background: leverage multi-sensor SNR

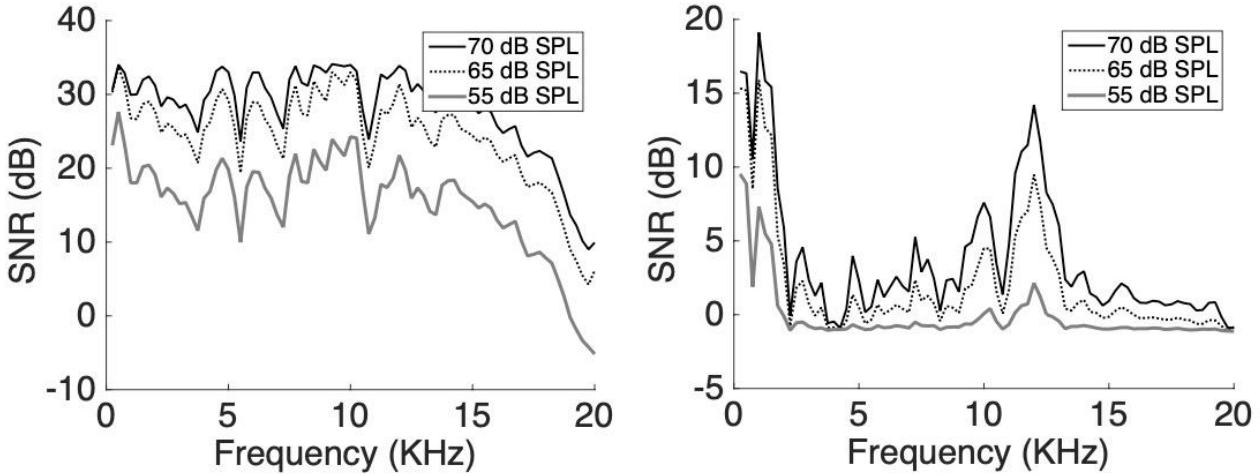


Figure 10: The SNR of (a) the microphone and (b) the vibra-motor at various frequencies for varying sound pressure levels (dB SPL). Note the unequal Y axis range.

Source: Listening through a Vibration Motor (*MobiSys'16)*

Past research has shown that different device components have different Signal-to-Noise Ratio (SNR) sensitivity.

Mic sensitivity depends heavily on weight and stiffness of its diaphragm Vibro-motor as sensor yields low sensitivity.

These readings can be combined to jointly create a more accurate signal.

Background: leverage combined sensor data

Sensor	Description
Accelerometer	Determines 3D acceleration (m/s^2 in x,y,z) --> mainly used to determine movement of the user
Gyroscope	Determine angular movement in a smartphone (radians/s) --> helpful for determining heading direction of a user
Magnetometer	Measure ambient geomagnetic field (magnetic values in x,y,z) --> used to determine the North, improve accuracy of gyroscope readings, and other metrics such as inclination, declination, total horizontal intensity, and total magnetic intensity
Barometer	Measure air pressure
Light sensor	Senses environment light where the smartphone is currently operating --> consists of Ambient Light Sensing (ALS), proximity sensors, IR LED
Camera	Smartphones usually use CMOS Backside Illumination (BSI), High Dynamic Range (HDR), Extended Depth of Focus (eDoF), and Near IR Capability (NIR)

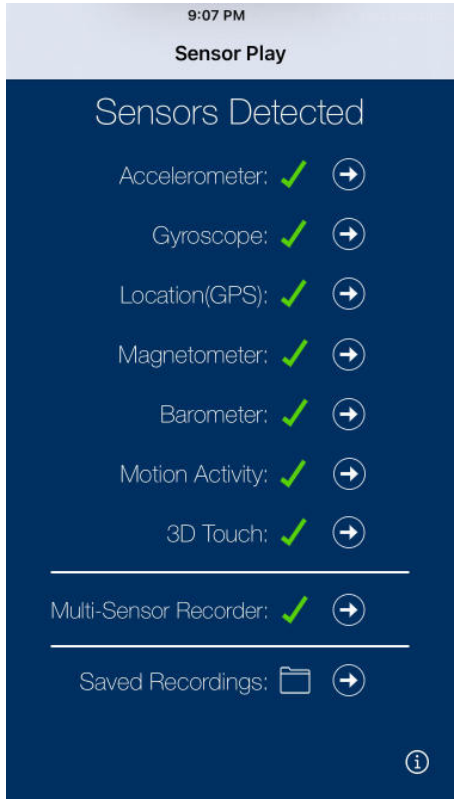
Source: Smartphone Sensor Based Indoor Positioning: Current Status, Opportunities, and Future Challenges (electronics ‘20)

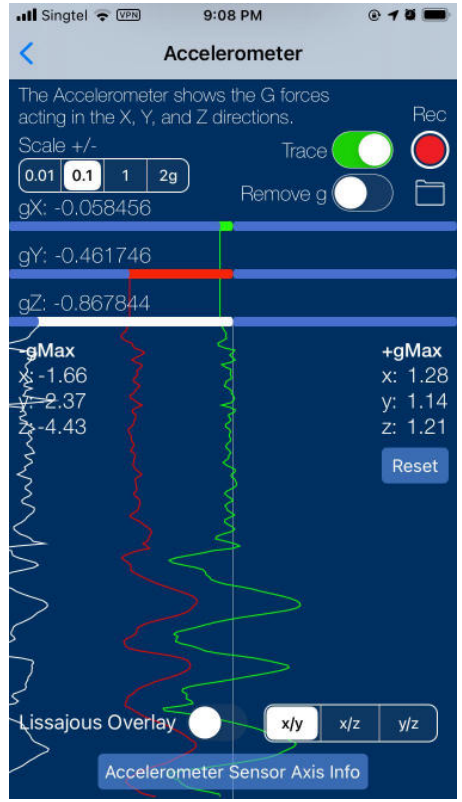
Methodology

Data Collection



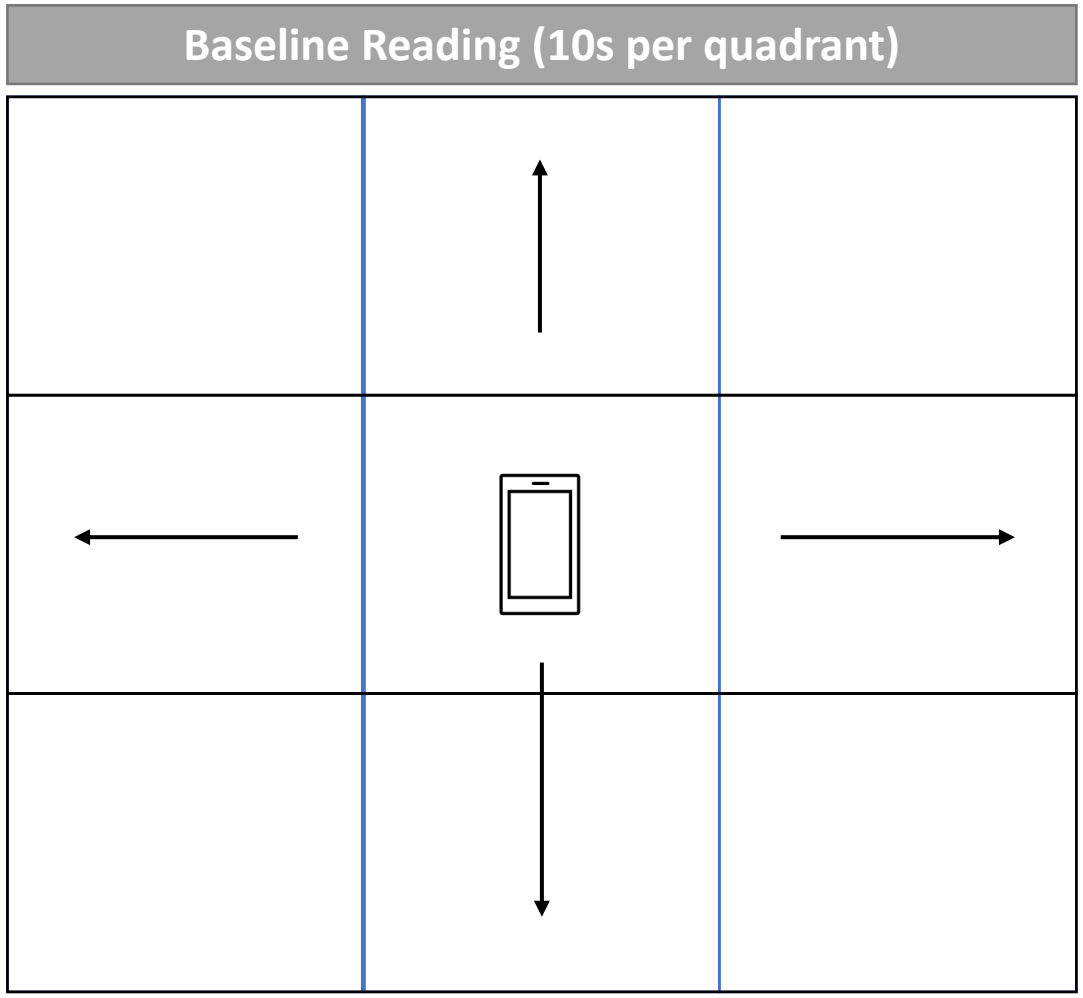
Source: Smartphone Sensor Based Indoor Positioning: Current Status, Opportunities, and Future Challenges (electronics ‘20)

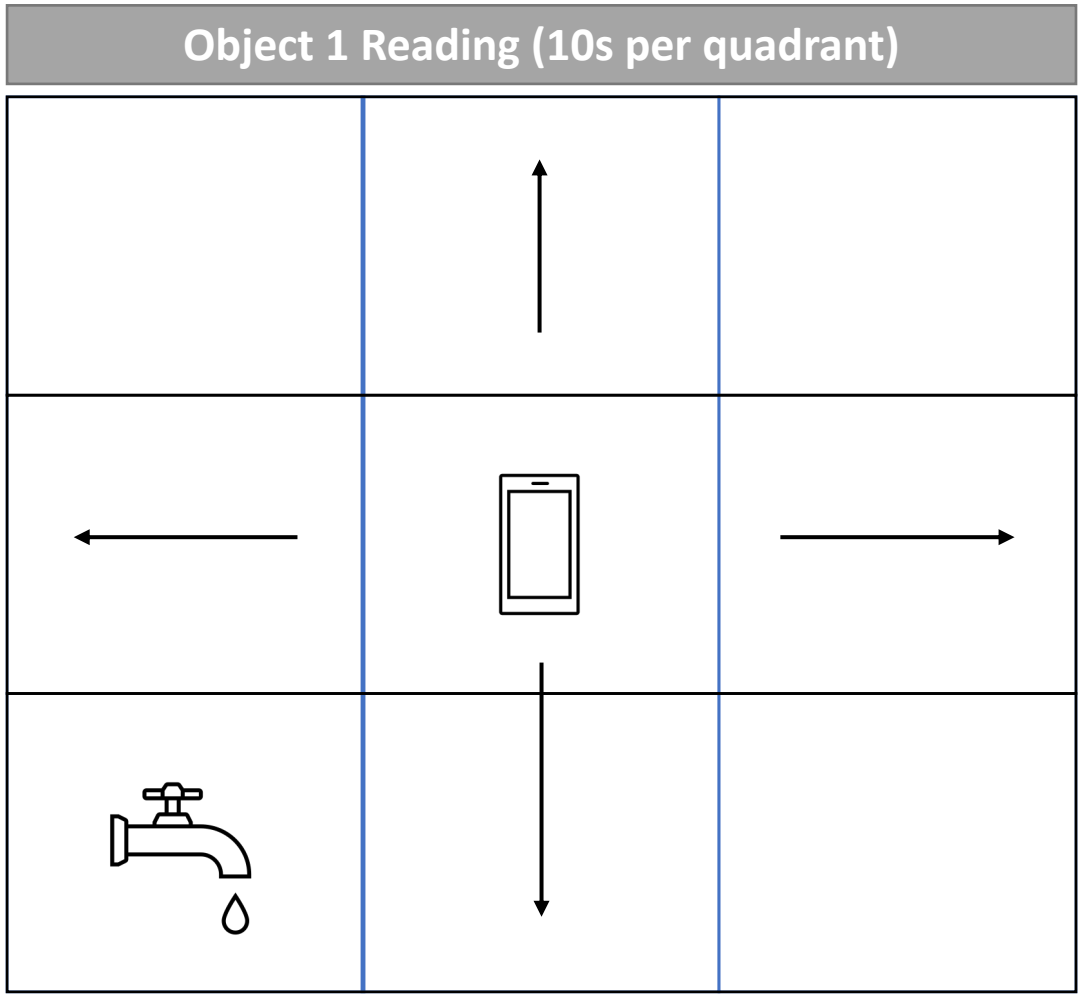




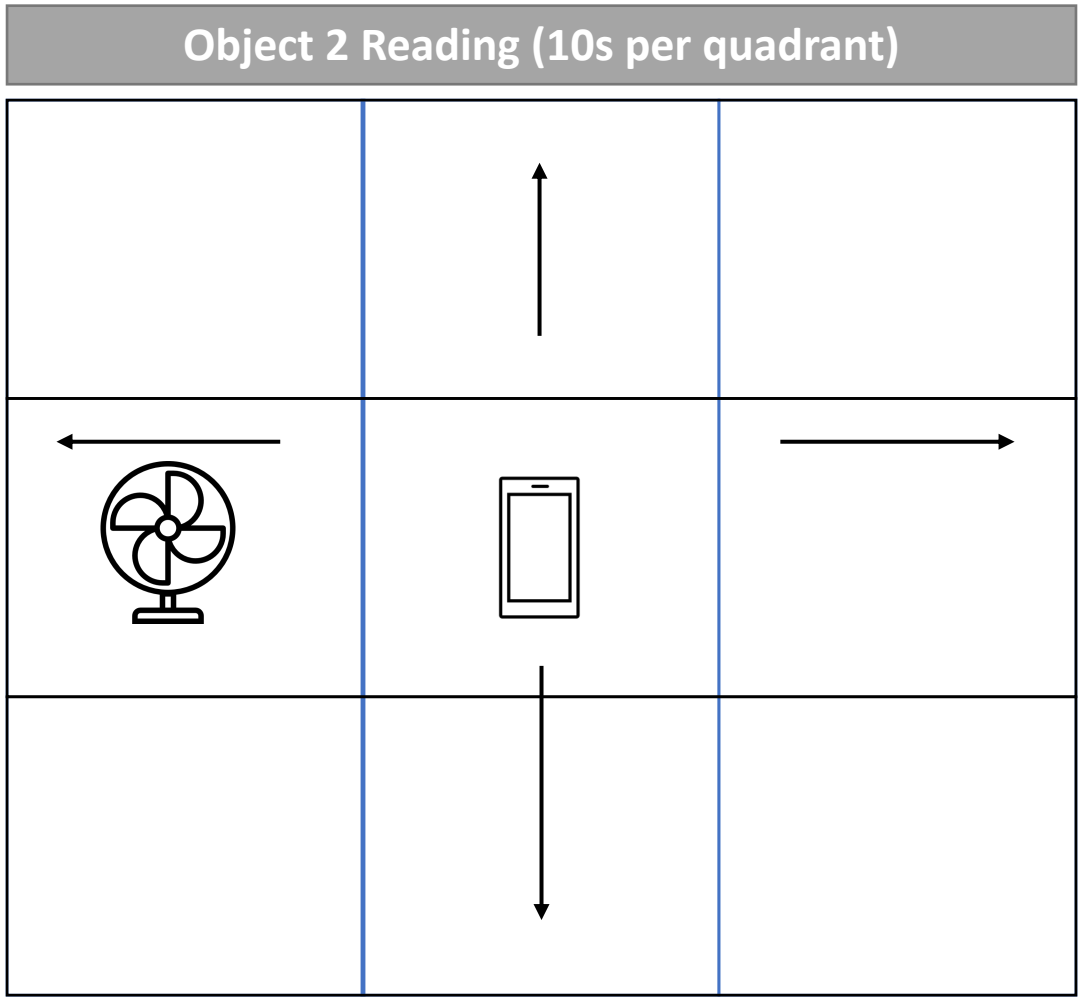
Source: Sensor Play app for the iPhone can record raw data for iPhone’s sensors

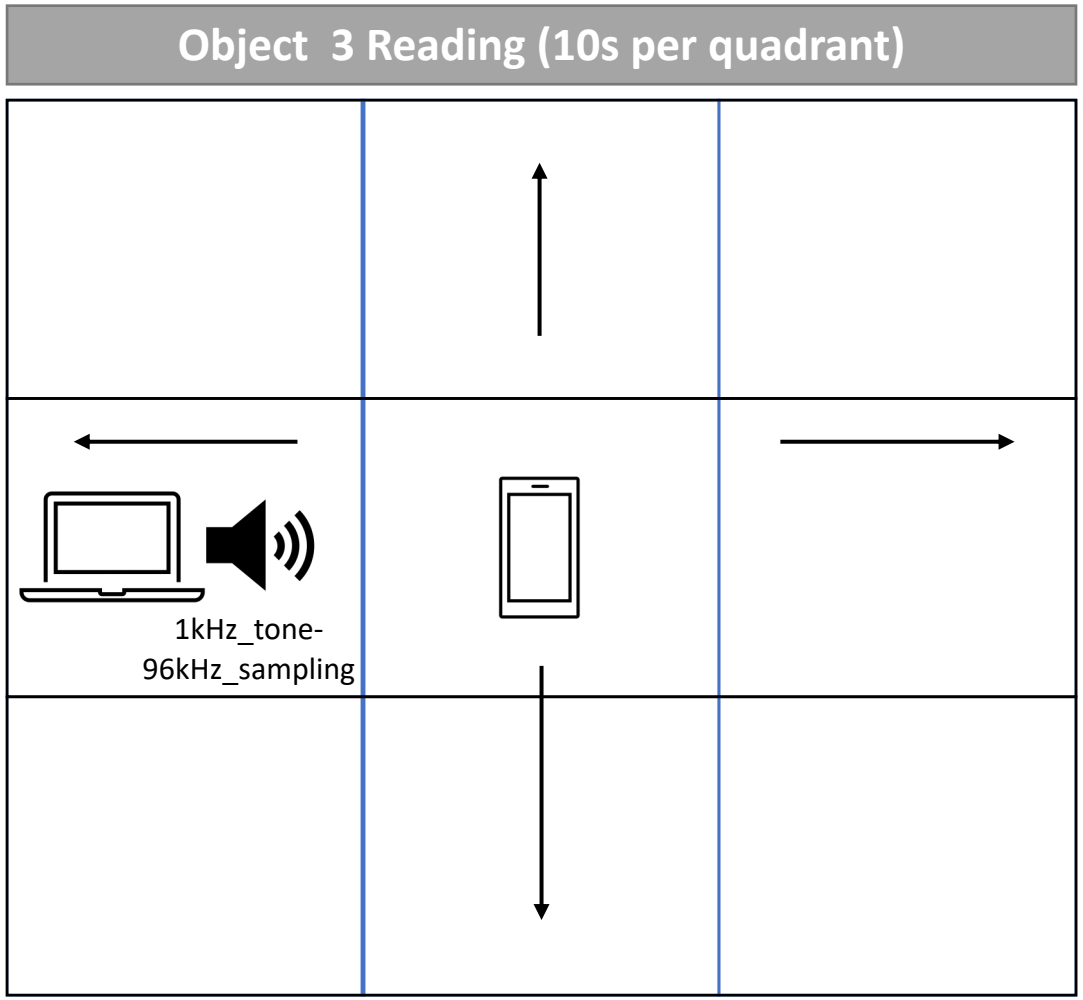
Data Acquisition: 9 Nodes And Three Objects





Data Acquisition: 9 Nodes And Three Objects





Data Acquisition: Dataset

Grouping	Dataset Number	Name	Data Frames (files)	Entries (rows)	Attributes (columns)
Training	001	Baseline	9	900	45
	002	Water Tap	9	900	45
	003	Fan	9	900	45
	004	Computer Audio	9	900	45
Testing	005	Baseline	1	900	45
	006	Water Tap	1	900	45
	007	Fan	1	900	45
	008	Computer Audio	1	900	45

Data Acquisition: Organization

Notebook	Name	Input Datasets	Techniques	Outputs
1	Exploratory Data Analysis	001 to 004	N/A	3D plots of sensor data
2	Supervised Learning: Regression Models	001 to 008	Linear Regression Ridge Regression Lasso Regression ElasticNet Regression KNN Regression	Regression Metrics: R^2 scores
3	Supervised Learning: Classification Models	001 to 008	Logistic Regression Random Forest Classification KNeighbors Classification DecisionTree Classification	Classification Metrics: F1 scores
4	Unsupervised Learning: Clustering Models	001 to 008	K-Means DBSCAN Agglomerative Clustering	Clustering Metrics: Silhouette scores

Experiments & Results

Exploratory Data Analysis

In [4]:
```
df2 = pd.read_csv('.././2021-03-29 Experiment Data/Sensor data 4 - center tap running.csv', sep=',', index_col='Timesta

df2.head()
```

Out[4]:

Timestamp	accelX(g)	accelY(g)	accelZ(g)	accelUserX(g)	accelUserY(g)	accelUserZ(g)	gyroX(rad/s)	gyroY(rad/s)	gyroZ(rad/s)	Roll(rads)	...	ActivityType	Act
29-Mar-2021 08:29:19.021	0.0380	-0.025	-0.9917	-0.0005	-0.0002	0.0224	-0.0013	-0.0201	-0.0112	0.0404	...	Stationary	
29-Mar-2021 08:29:19.024	0.0380	-0.025	-0.9917	-0.0005	-0.0002	0.0224	-0.0013	-0.0201	-0.0112	0.0404	...	Stationary	
29-Mar-2021 08:29:19.027	0.0380	-0.025	-0.9917	-0.0022	-0.0048	-0.0232	-0.0294	-0.0140	-0.0091	0.0403	...	Stationary	
29-Mar-2021 08:29:19.030	0.0401	-0.032	-1.0382	-0.0022	-0.0049	-0.0232	-0.0374	-0.0121	-0.0098	0.0403	...	Stationary	
29-Mar-2021 08:29:19.040	0.0406	-0.026	-1.0572	0.0015	-0.0039	-0.0562	-0.0127	-0.0002	-0.0069	0.0403	...	Stationary	

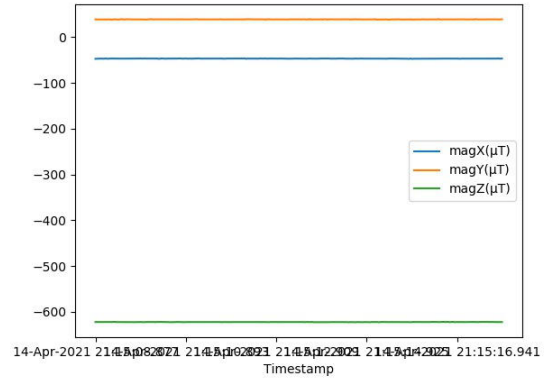
5 rows × 43 columns

In [5]:
```
num_points, num_attributes = df.shape

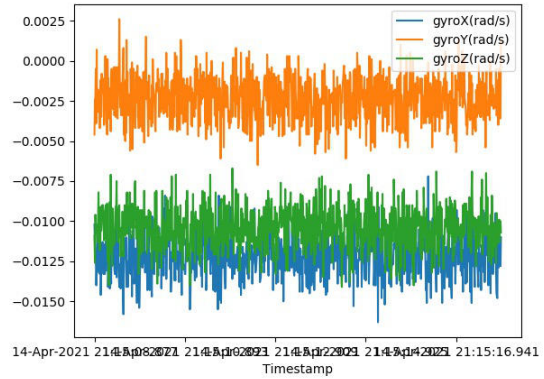
print("There are {} data points, each with {} attributes.". format(num_points, num_attributes))
```

```
There are 1085 data points, each with 43 attributes.
```

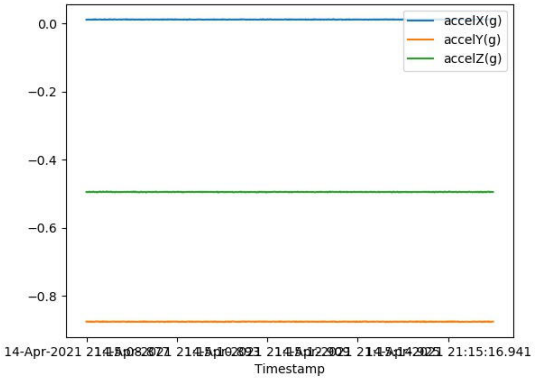
Sensor Readings (Baseline)



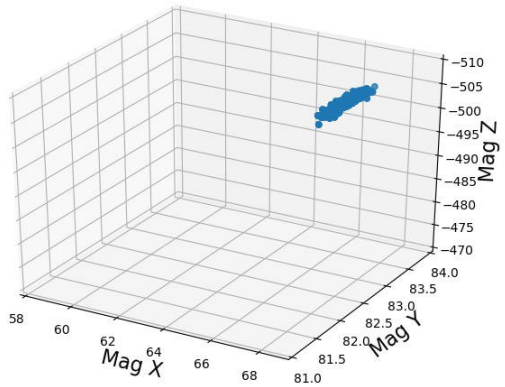
Magnetometer

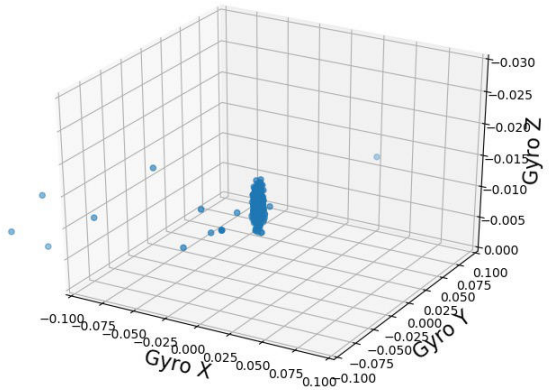


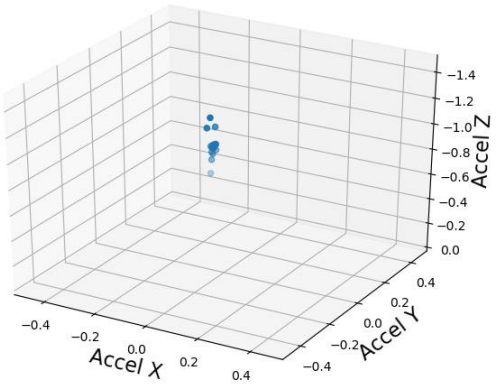
Gyroscope



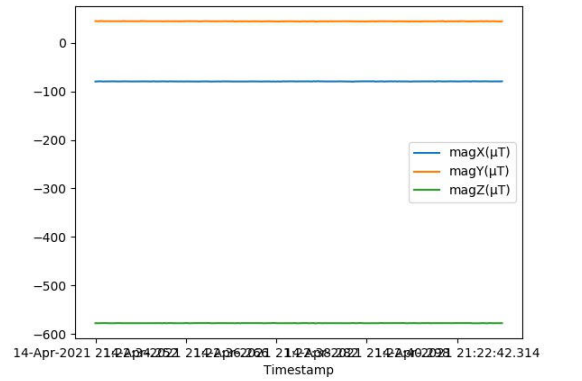
Accelerometer



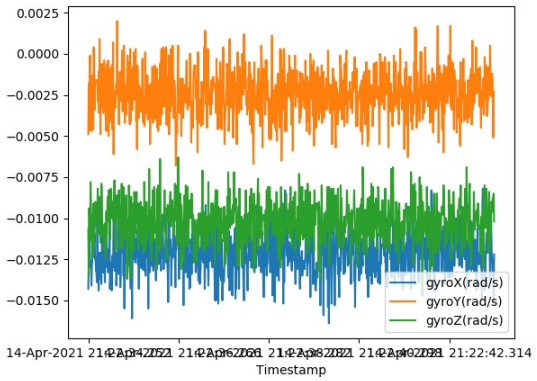




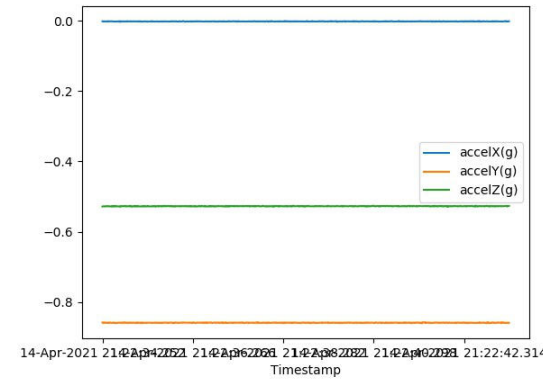
Sensor Readings (Water Tap Running)



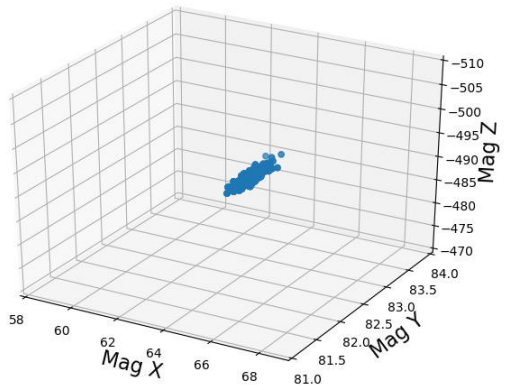
Magnetometer

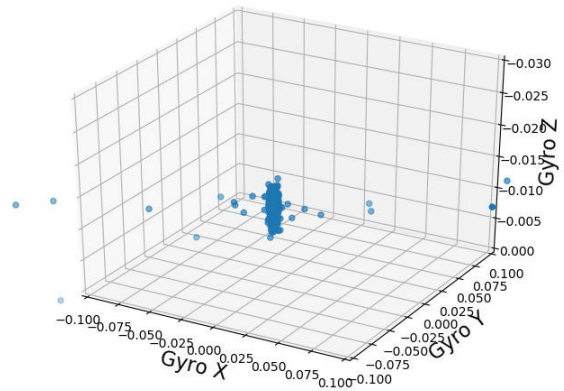


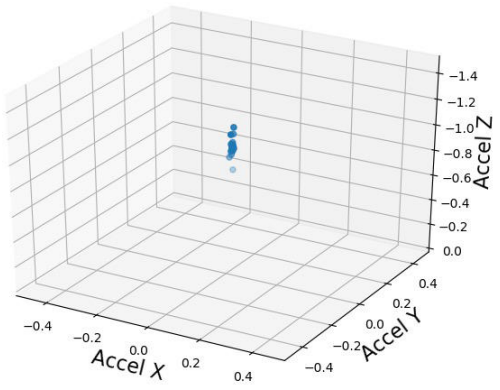
Gyroscope



Accelerometer





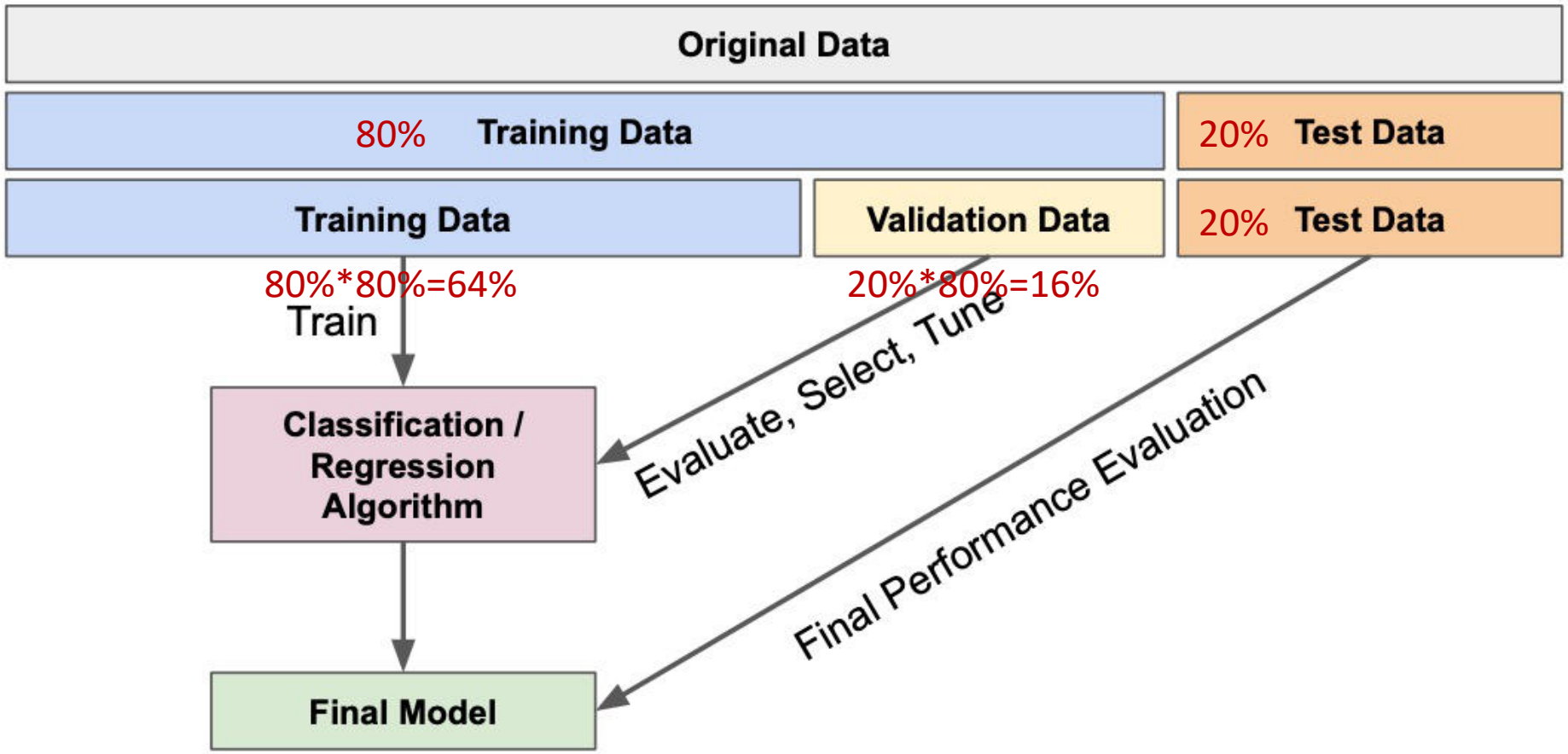


Process

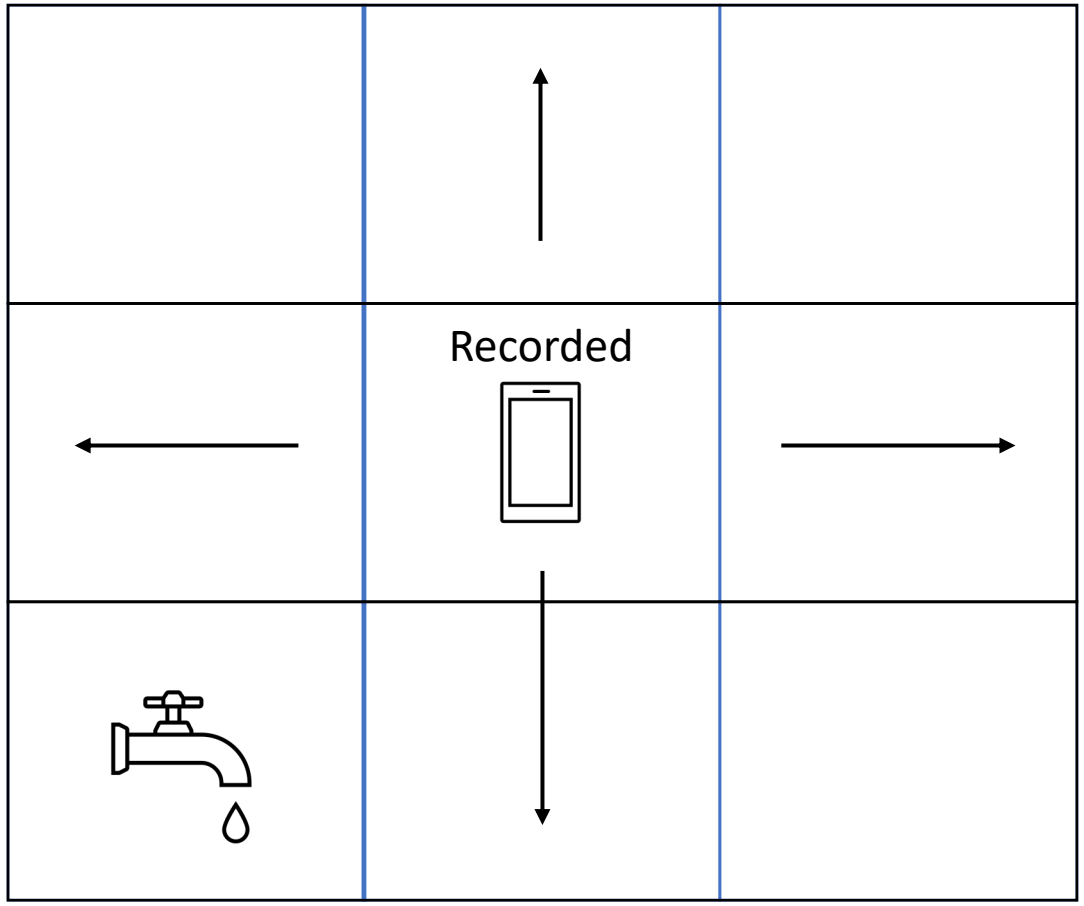
- Choose libraries:
  - Numpy: python arrays
    : python data structures and analysis
    : plotting graphs
  - Librosa: audio processing
  - Scikit-learn: data mining and machine learning
- Other libraries: Keras, PyTorch, TensorFlow
- Data Management:
  - Import Data
  - Clean Data
  - Transform Data
  - Split up the data for training, validation, and testing

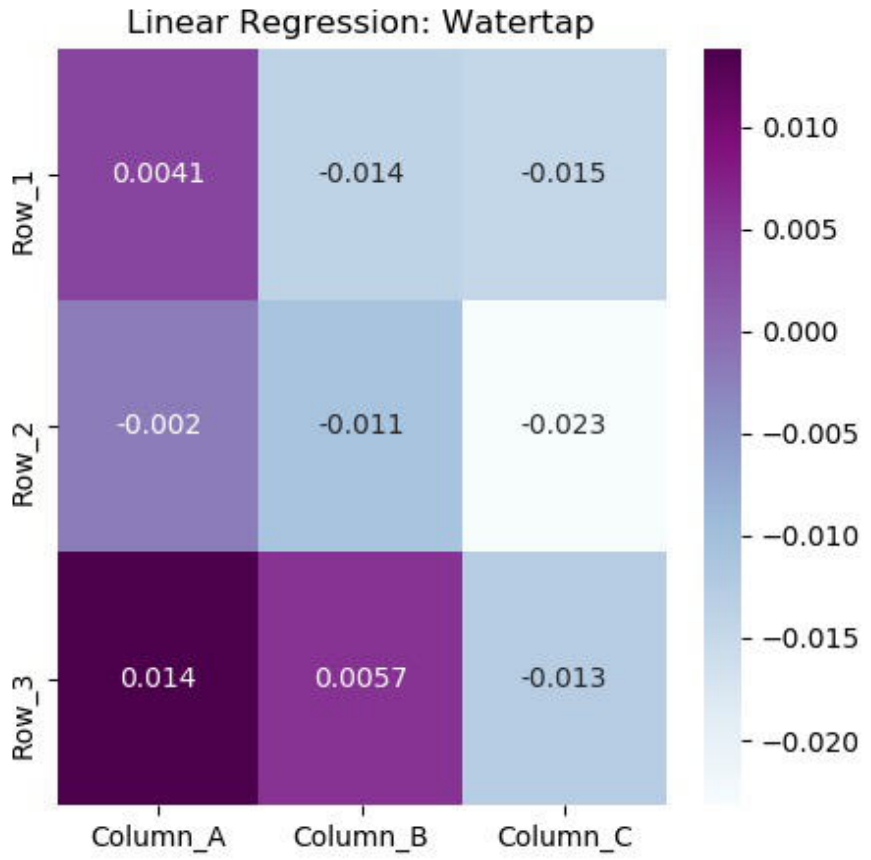
Present results: Heatmap

Data Splitting



Linear Regression: Water Tap

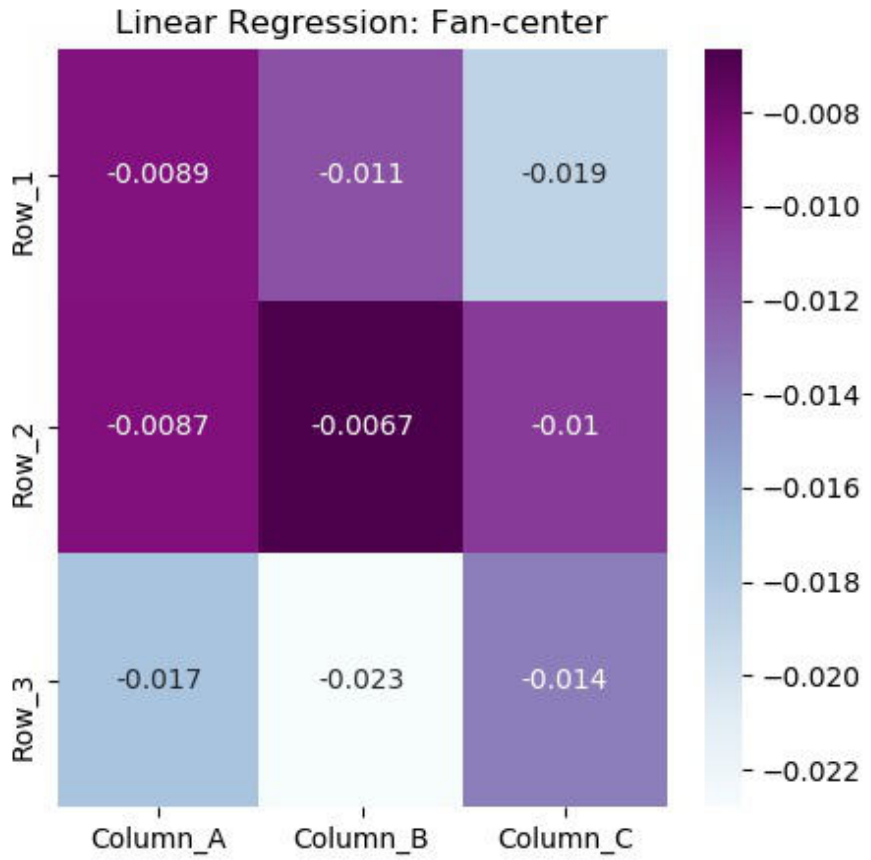


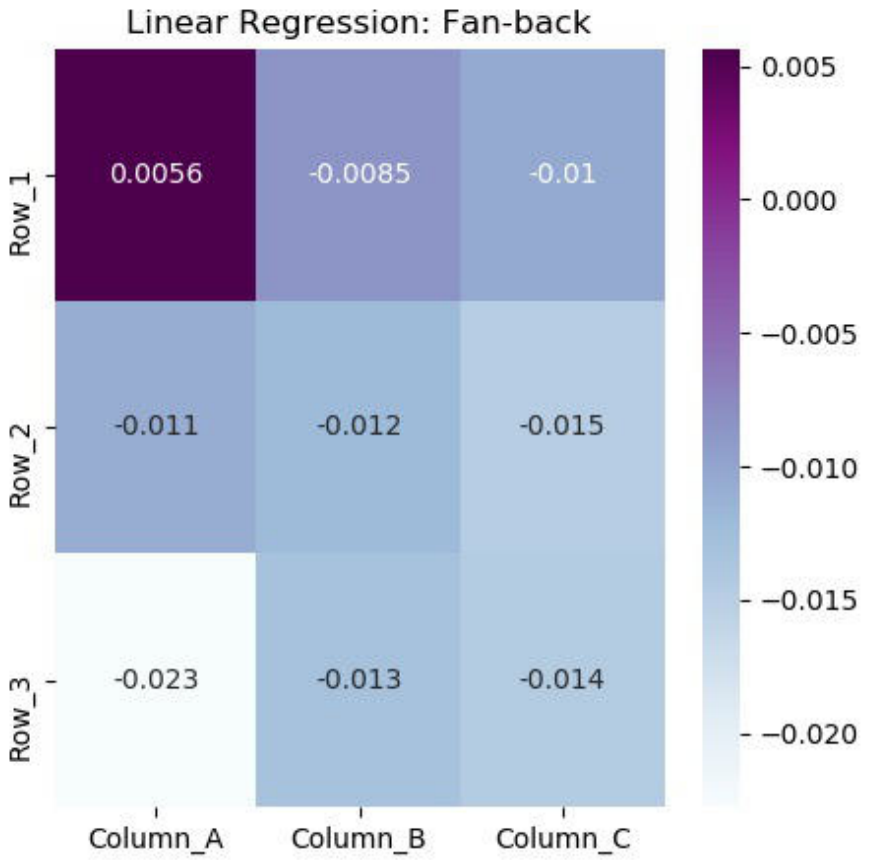


Outcomes

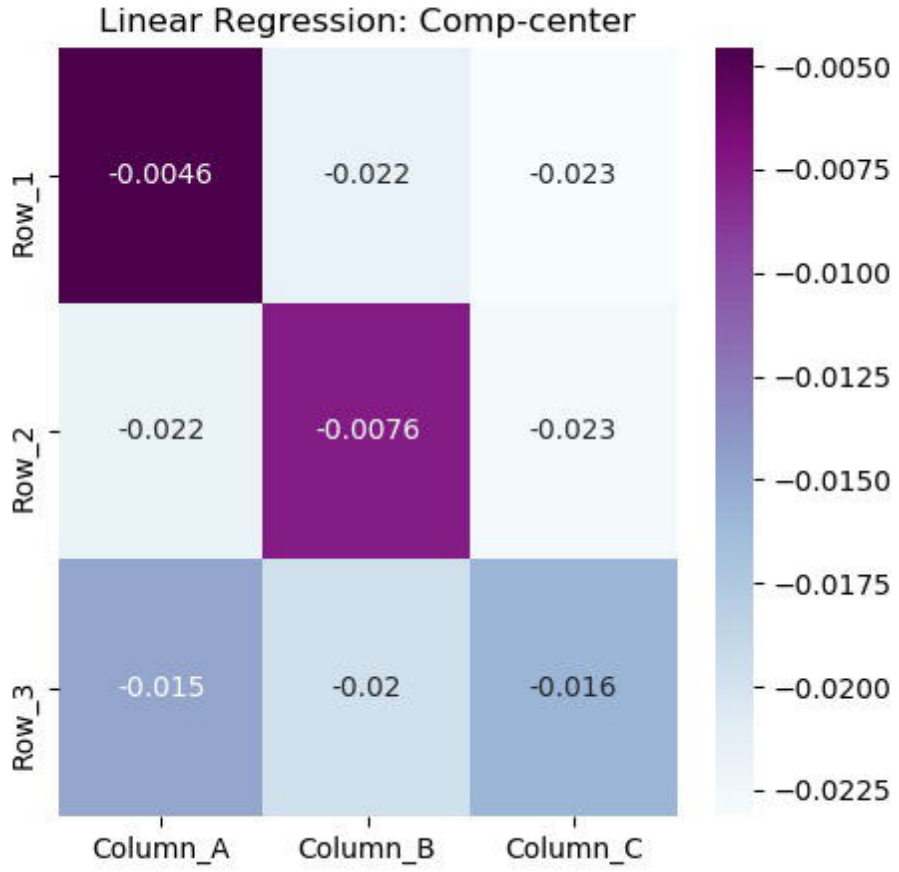
Regression Method	Fan	Computer Audio	Water	Total
Linear	2/2	2/2	1/2	5/6
Ridge	0/2	0/2	0/2	0/6
Lasso	2/2	0/2	0/2	2/6
ElasticNet	0/2	0/2	0/2	0/6
KNN	2/2	0/2	1/2	3/6

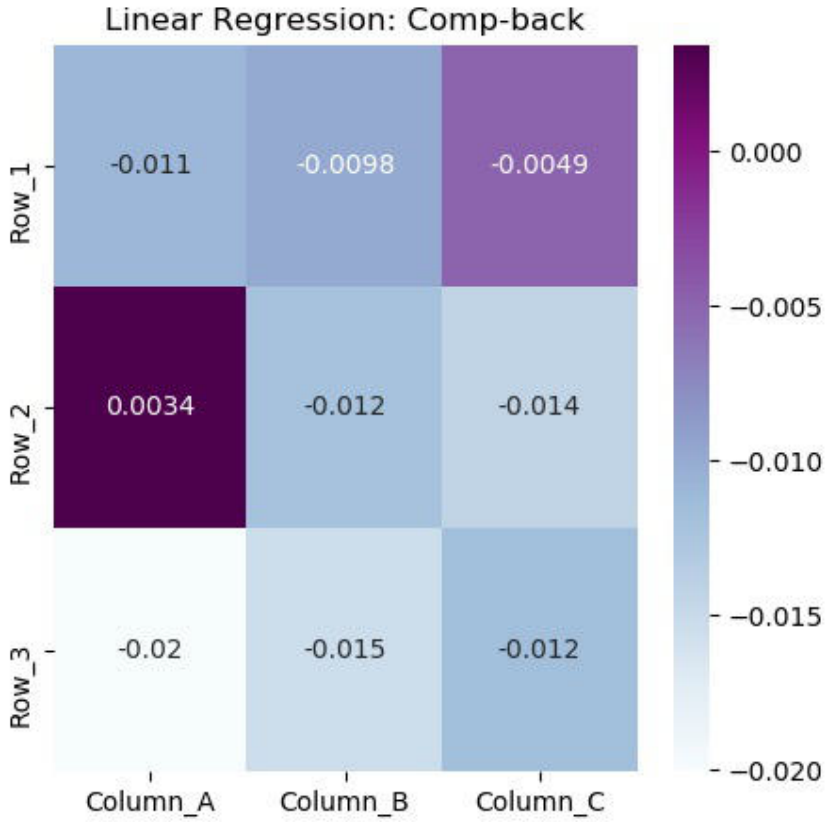
Linear Regression: Fan



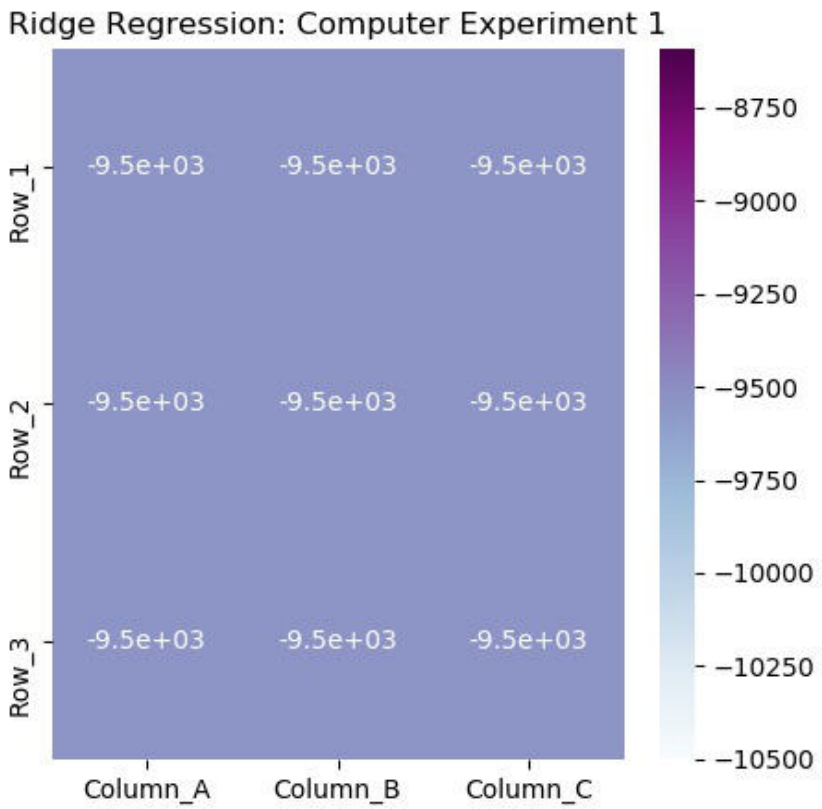


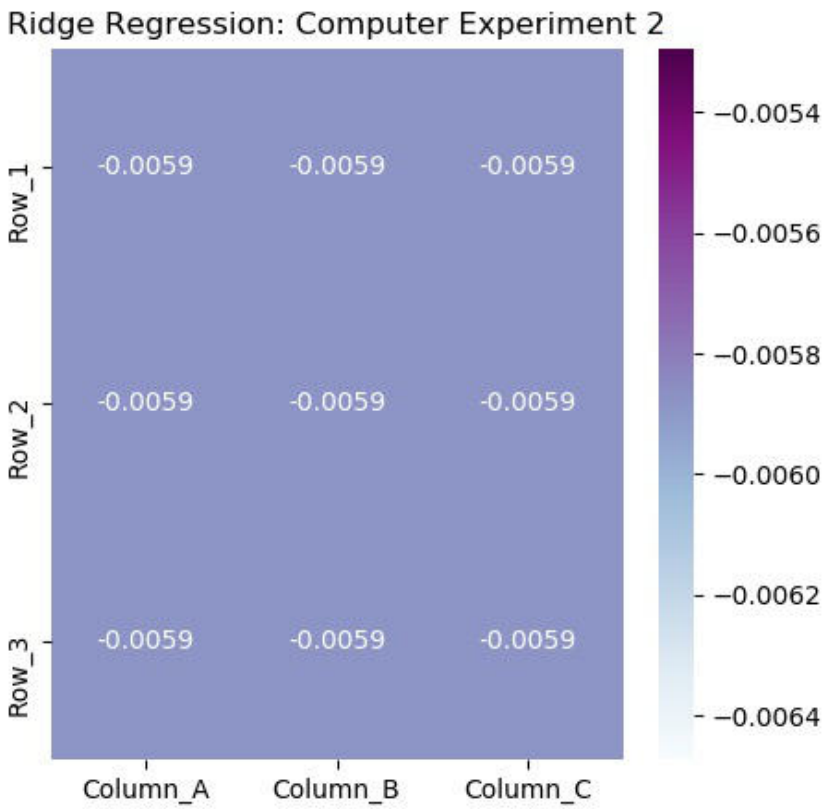
Linear Regression: Comp



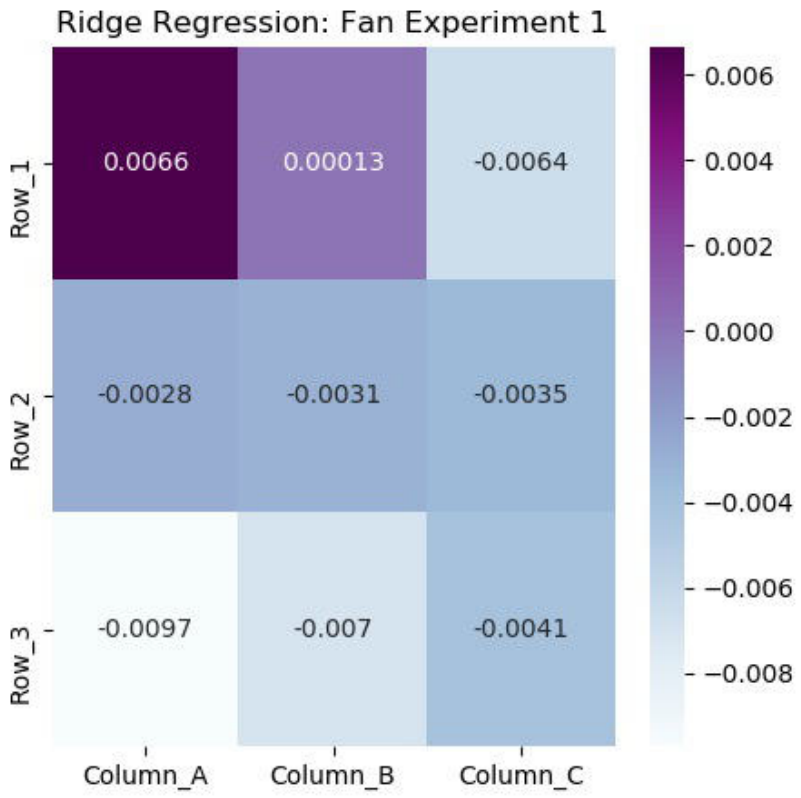


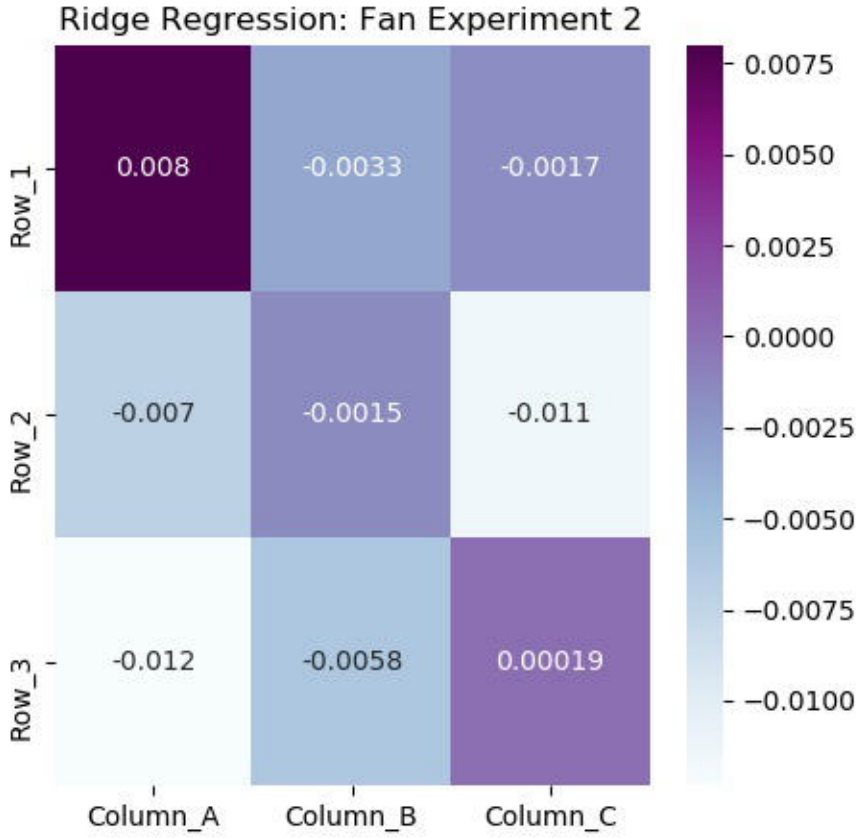
Ridge Regression: Comp Audio



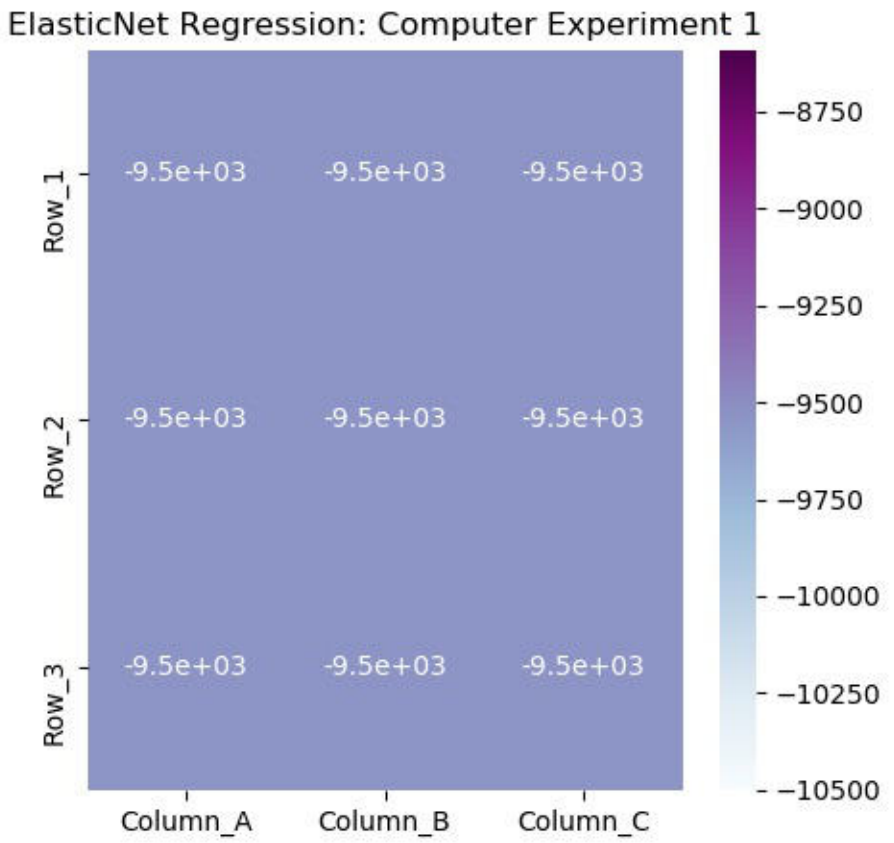


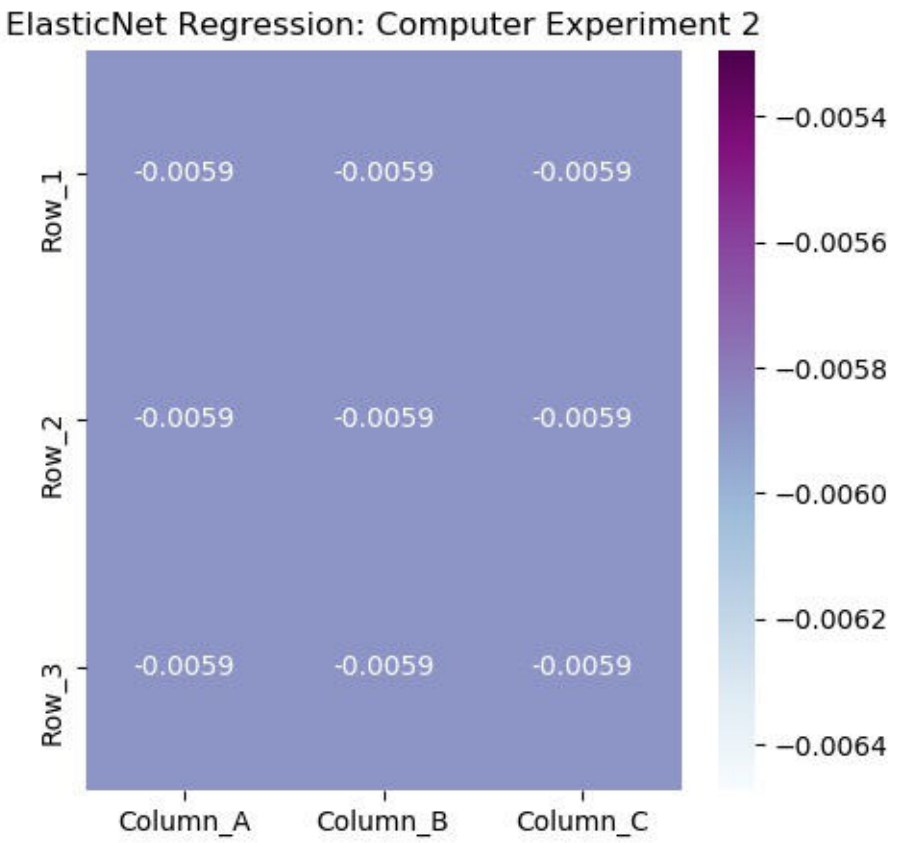
Ridge Regression: Fan



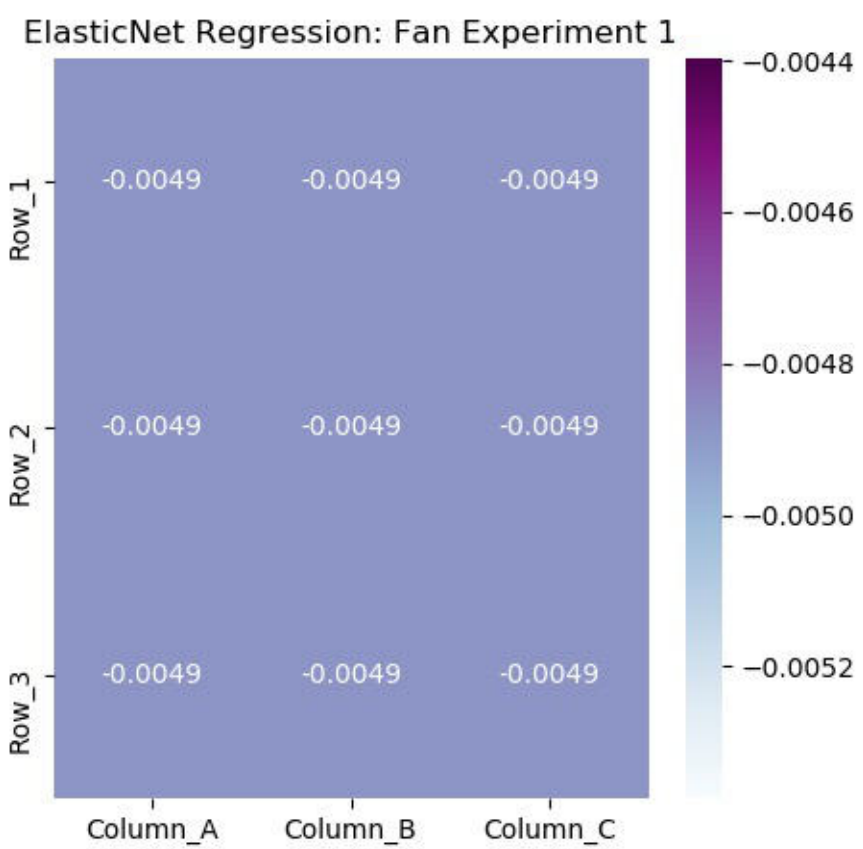


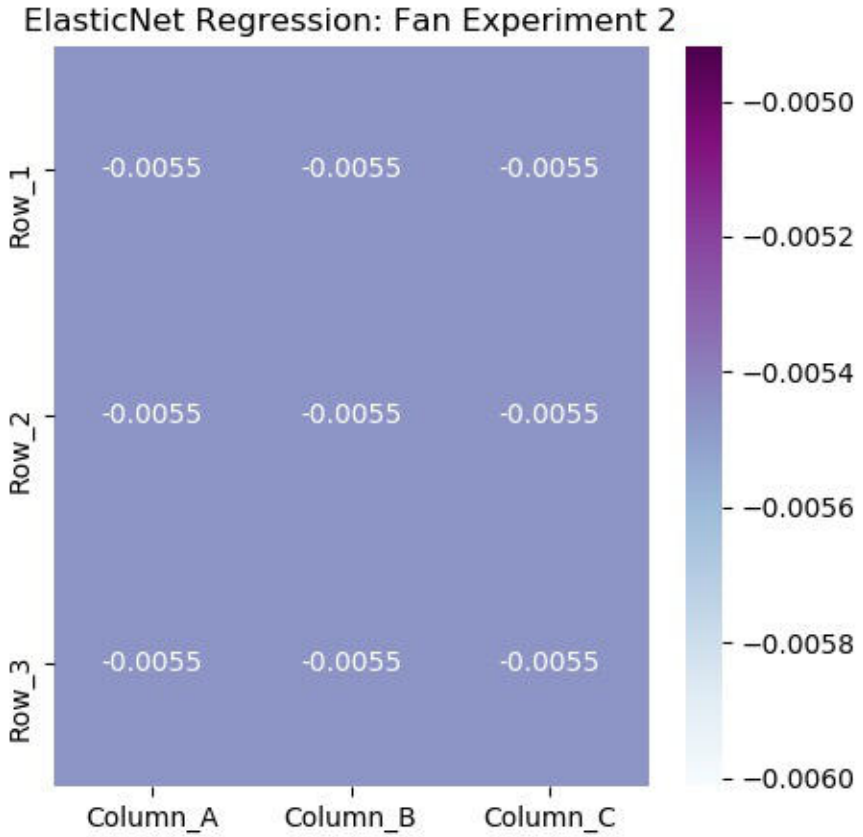
ElasticNet Regression: Comp Audio





ElasticNet Regression: Fan





Limitations and Future Work

Limitations

1. Background noise: acoustic, EMR, vibrations, etc.
2. Sensor readings not sensitive enough to pick up certain objects at a distance
3. User was in the room when sensor data was recorded
4. Same device used to imitate different nodes (i.e., data not recorded at the same time)

Future work

Audio Signal	•Repeat experiments using mic recordings
Supervised learning: classification methods	•Logistic Regression •Random Forest Classification •KNeighbors Classification •DecisionTree Classification
Unsupervised learning: clustering methods	•K-Means •DBSCAN •AgglomerativeClustering
Hyper-parameter tuning	•Grid Search Cross Validation
Additional experiments	•More data collection (larger sample size) •Less-noisy room •Different objects •More granular matrix (e.g., 3 x 3 → 5 x 5)

Lessons Learned

1. Experimenting with sensors and other hardware takes time to properly design and perform (multiple iterations will be needed)
2. Data cleaning can be simplified or avoided with good data collection
3. Data fusion can be re-framed as hyper-parameter tuning using number of features as the parameter
4. Applying unsupervised machine learning techniques, such as clustering, requires modifying the IoT data significantly
5. Applying some supervised machine learning techniques, such as classification, also requires modifying the IoT data significantly

Thank you

Related Papers

[1]	, and , "A Survey of Indoor Localization Systems and Technologies," IEEE, 2019.
[2]	and , "Sound based localization and identification in industrial environments," IEEE, 2017.
[3]	, , , , , and , "Structural Health Monitoring: An IoT Sensor System for Structural Damage Indicator Evaluation," Sensors, 2020.
[4]	, B. Fan, and , "Fault detection and diagnosis for buildings and HVAC systems using combined neural networks and subtractive clustering analysis," 2013.
[5]	and , "RADAR: An In-Building RF-based User Location and Tracking System," IEEE, 2000.
[6]	T. de Haan, "BLE Localization Using Switched-beam Angle Of Arrival: For Pallet Localization In Warehouses," 2018.
[7]	E. Asimakopoulou and , "Buildings and crowds: Forming smart cities for more effective disaster management," IEEE, 2011.
[8]	, and , "Smart City and the Applications," 2020.
[9]	D. Snoonian, "Smart Buildings," IEEE, 2003.
[10]	, "Demand response and smart grids — A survey," 2014.
[11]	and , "Smartphone Sensor Based Indoor Positioning: Current Status, Opportunities, and Future Challenges," Electronics, 2020.